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SUMMARY

Computer Science student with hands-on experience building real-world projects, including data systems and real-time applications. I focus on understanding how systems work end-to-end and solving practical problems through building. I enjoy breaking down complex problems, working across both frontend and backend, and turning ideas into simple, working solutions

EDUCATION

Bachelor of Engineering – Computer Science & Engineering	2023-Present	CGPA:9.58
<i>Mangalore Institute of Technology & Engineering</i>		
Senior Secondary (12th) - KSEEB	2021-2023	Percentage:96.1%
Secondary School (SSLC) - KSEEB	2020-2021	Percentage : 94.2%

SKILLS

Languages	: Java, Python, C
Backend	:Django , REST APIs
Frontend	: Flutter (BLoC, MVVM)
Database	: PostgreSQL, MongoDB, Firebase
Cloud	: AWS (EC2, S3, VPC, IAM)
Tools	: Git, GitHub, VS code
Core Concepts	:Data Structures & Algorithms, OOP, System Design Basics

INTERNSHIP

Mindstack Technologies Pvt Ltd Flutter Developer	August 2025 - October 2025(Hybrid)
- Built and enhanced cross-platform mobile app features using Flutter with BLoC architecture for scalable state management. - Integrated REST APIs with proper error handling and maintained clean coding standards. - Maintained clean architecture principles and documented API contracts	

PROJECTS

Temporal Differential Reachability Analysis for Detecting Change-Induced Security Risk in Evolving Java Codebase	Group of 4	Present
Technologies: Java, Python, AST Analysis, Call Graphs, NetworkX, Git		
Developing a static analysis framework to detect security risks introduced by code changes in Java codebases. Performs differential analysis across versions to identify new execution paths and potential vulnerabilities. Utilizes call graph analysis to prioritize high-risk execution paths.		
Federated Query Engine (Multi-Database Query System) 2026		
Technologies: Python, PostgreSQL, MongoDB, HTML, CSS, Javascript		
Built a federated query engine to execute unified queries across SQL and NoSQL databases. Designed a modular pipeline with query parsing, planning, and execution components. Implemented schema extraction and entity		

detection to handle ambiguous queries and deployed the system on AWS using multi – tier setup with EC2,RDS,MongoDB, Atlas and S3.

One-to-One Video Calling Application | 2025

Technologies: Flutter, Django, PostgreSQL, Agora SDK, BLoC

Developed a real-time video calling application with secure peer-to-peer communication. Integrated Agora SDK for low-latency streaming and implemented BLoC architecture for scalable state management. Designed backend APIs for authentication and session handling. Ensured reliability by handling token lifecycle, retries, and error handling for cases like timeouts.

BlinkEasy - Blink Detection Application | 2025

Technologies: Flutter, MediaPipe, Flask, Firebase

Built a computer vision-based application to monitor blink rate and detect eye strain from prolonged screen usage. Integrated MediaPipe with a Flask backend for real-time facial landmark detection and compute blink frequency and added reminder system based on the eye strain. Also implemented user authentication and data storage using Firebase, while handling camera permissions and continuous video processing on the frontend.

COURSES & WORKSHOPS

- Mastering System Design (Udemy) – Distributed systems, scalability patterns, architecture trade-offs
- AWS cloud fundamentals (Skill Builder) - EC2, S3, VPC, IAM, RDS, Cloud front fundamentals
- Generative AI – IUCEE
- Java Spring Boot & REST APIs Workshop – Backend development & API integration

ACHIEVEMENTS & ACTIVITIES

- Collaborated in a 6-member team to organize a college-level technical event with 90+ teams; designed coding and logic-based challenges
- Presented an AI-based gesture recognition solution at SAP Hackfest 2025 for assisting deaf individuals
- Competed in Prompt Wars LLM Competition, focusing on prompt engineering and model evaluation
- Participated in technical events including TCS TechBytes (regional round) and various inter- and intra-college competitions, exploring emerging technologies.

HOBBIES

Technical Blogging:

- Simplifying complex computer science concepts using relatable analogies and storytelling. Focused on making technical topics easy to understand through clear and engaging explanations
- Published 12+ articles with 50+ subscribers by breaking down complex ideas into clear and relatable explanations
- Cover topics such as system behavior, operating system concepts, and everyday technology (e.g., disk bottlenecks, clipboard internals, location tracking, image formats)